Database Lab : Replication Lag Telemetry Table & Monitoring Setup

Duration : 30 Min.

Scenario

As a Database Operations Engineer at NIIT, reporting replicas are experiencing lag under burst write traffic. You must establish a replication telemetry audit database and table in MySQL, recording replication metrics, lag seconds, and escalation flags.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **MySQL Service:** `localhost:3306` | DB: `mysql` | User: `root` | Password: `MySQLPassword123!` | CLI: `mysql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **MySQL Monitoring Database:** `<prefix>_rep_monitor` - **Replication Log Table:** `replica_telemetry`

Task Objectives

Perform the following configuration steps directly via `mysql` CLI:

1. Create Monitoring Database

- In MySQL, create database `<prefix>_rep_monitor`.

2. Create Replica Telemetry Table

- Inside `<prefix>_rep_monitor`, create table `replica_telemetry` with columns:
- `replica_id VARCHAR(50) PRIMARY KEY`
- `channel_name VARCHAR(50) NOT NULL`
- `seconds_behind_primary INT NOT NULL`
- `replication_state VARCHAR(30) NOT NULL`
- `escalation_required BOOLEAN NOT NULL`

3. Insert Telemetry Records

- Insert two simulated replica status entries:
- `('rep-analytics-01', 'main_channel', 2, 'STREAMING', FALSE)`
- `('rep-reporting-02', 'reporting_channel', 1840, 'STALLED', TRUE)`

Verification

Once completed, run the verification script:

```
python run.py
```

The driver connects to MySQL on `localhost:3306` and queries `<prefix>_rep_monitor.replica_telemetry`.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	MySQL Database (_rep_monitor) Creation	4 Marks
TC2	Table (replica_telemetry) Schema & Column Constraints	3 Marks
TC3	Replication Metric Data Population & Escalation Logic	3 Marks

Total Score: 10 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- All commands must be executed in the live MySQL console.